Rank-Deadline Coverage for Risk-Limited Kinase Counter-Screening: A Cross-Panel Translation Audit

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Abstract

Diversity-aware assay ordering can find kinase-profile activity missed by ranking each target independently, but it can also postpone a high-probability assay arbitrarily far. We introduce *rank-deadline coverage*: greedy weighted profile coverage constrained so that every kinase stays within K positions of a chemical-similarity-weighted marginal order. The method gives a deterministic first-hit delay bound, requires no training, and runs on a CPU. With $K = 9$, it retained 80.6% of an unconstrained coverage gain under PKIS2 chemotype exclusion ($+0.0433$ discovery-curve area, 95% CI $[0.0321, 0.0545]$), while all eight development-condition estimates versus marginal ranking were nonnegative. We then froze the method and tested a separate PKIS1 panel after removing eight structures shared with development. Mean gains did not transfer: the external $> 80\%$ effect was -0.0206 ($[-0.0306, -0.0117]$). The guarantee did transfer. Across 4,428 held-out compound–condition evaluations, bounded coverage caused zero delays of ten or more assays, versus 163 for unconstrained coverage; on external PKIS1 it was also consistently less harmful than unconstrained coverage. Thus the method solves a real tail-risk problem, not universal ranking superiority. The untouched external failure delineates when retrospective diversity gains are not evidence of translatable counter-screen benefit.

1 Introduction

Kinase inhibitors often exhibit polypharmacology, so broad biochemical or binding panels are central to selectivity characterization [4, 6, 7]. When only a small counter-screen panel can be run initially, the operational question is sequential: for a new compound, which kinase assay should be run next to discover the first recorded profile activity? This is active search rather than full-matrix prediction [5]. A marginal ranker tests kinases with the largest estimated activity probability. A profile-coverage ranker instead avoids assays that hit the same reference compounds, potentially exposing chemically distinct liability patterns sooner. The latter is a monotone submodular-coverage idea [1, 8], but unconstrained diversification can move an individually strong assay arbitrarily far down the queue.

This tradeoff is consequential in drug discovery: an average gain cannot justify a policy that occasionally spends many extra experiments before finding a liability. We therefore ask whether profile diversity can be used inside an explicit marginal-rank trust region. We contribute:

1. a deterministic rank-deadline algorithm with a per-compound bound on the delay to first activity relative to marginal ranking;
2. frozen evaluation on three public kinase-profile sources, 1,107 panel-specific compound episodes (1,102 unique parent structures), and chemical-series exclusions;
3. an external PKIS1 test fixed only after method development, including the negative transfer result rather than selecting a favorable dataset; and

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4. complete CPU artifacts, hashes, provenance, and license-aware data construction at zero API cost.

The closest drug-discovery work uses bioactivity matrices to select informative *compounds* for a new target [11]. We reverse the decision axis—selecting target assays for a new compound—and add a hard displacement constraint. We do not claim new general submodular theory or that biochemical profile activity establishes cellular engagement, toxicity, efficacy, or clinical safety.

2 Rank-deadline profile coverage

For held-out compound h, references $i \in R_h$ have binary profiles y_{it} over candidate kinases $t \in T_h$. Missing assays remain missing. Morgan radius-2, 2,048-bit Tanimoto similarities give fixed weights $w_{hi} \propto \exp(8 s_{hi})$ [9]. Let p_{ht} be the $\mathrm{Beta}(1, 1)$-smoothed weighted activity frequency. MARGINAL sorts by p_{ht} with opaque target-key ties.

For a selected set S, weighted profile coverage is

$$F_h(S) = \sum_{i \in R_h} w_{hi} \mathbb{1}[\exists t \in S : y_{it} = 1] \,. \tag{1}$$

UNBOUNDED COVERAGE greedily maximizes the marginal gain in F_h, using p_{ht} and the opaque key for ties. This rewards assays that cover reference activity patterns not already covered by earlier assays.

Let $r(t)$ be target t's zero-based full marginal rank. At output position j, BOUNDED COVERAGE uses a fixed displacement K:

1. if any unselected target has deadline $r(t) + K \leq j$, select the one with smallest $r(t)$;
2. otherwise, maximize the current gain in Eq. 1 only among targets satisfying $r(t) \leq j + K$.

Ties follow the unbounded rule. We fix $K = 9$ from the declared unacceptable delay of ten assays, not from an outcome sweep.

Proposition 1 (First-hit backstop)**.** *Extend the bounded rule to a complete order. Every target moves at most K positions earlier or later than its marginal rank. Consequently, at any evaluation budget B, its censored cost to first positive is no more than K assays worse than marginal ranking.*

Proof. Eligibility directly prevents an advance beyond K. For lateness, by position j the earliest-deadline rule has selected every target with $r(t) + K \leq j$: induction adds the next marginal-rank deadline at each step, and selecting an eligible future target before a deadline creates no backlog. Thus displacement is at most K. If marginal's first positive rank r^* has $r^* + K < B$, it is tested by that position. Otherwise censoring bounded cost at $B + 1$ gives delay $B - r^* \leq K$. If marginal finds no positive by B, positive delay is impossible. □

This is an outcome-independent safety property. It does not prove that the reference coverage objective transfers to the held-out compound; our external experiment tests exactly that missing premise.

3 Frozen evaluation

Panels and biological endpoints. We never pool assay modalities. **Klaeger/ChEMBL 30** contains 111 clinical kinase-drug parent structures and 146 commonly measured kinases [7, 10]. A recorded binding liability is an exact K_d within tenfold of the compound's best measured documented-target K_d; documented targets are excluded from candidates. **PKIS2** contributes 640 parent structures and 392 parent kinases from 1 μM percent-inhibition profiles [2]; activity is strictly $> 90\%$, with $> 80\%$ sensitivity. **PKIS1 external** comes from the authors' archived raw Nanosyn 1 μM file [3, 11]. Repeat assays are averaged, constructs sharing a parent target are max-collapsed, salts are reduced to the largest organic fragment, and duplicate parent structures are median-collapsed. Eight structures present in PKIS2 and none present in Klaeger are removed, leaving 356 structures and 200 parent kinases. The reconstructed 366-by-224 pre-collapse matrix matches the authors' archive to 1.42×10^{-14} maximum absolute error with identical missingness.

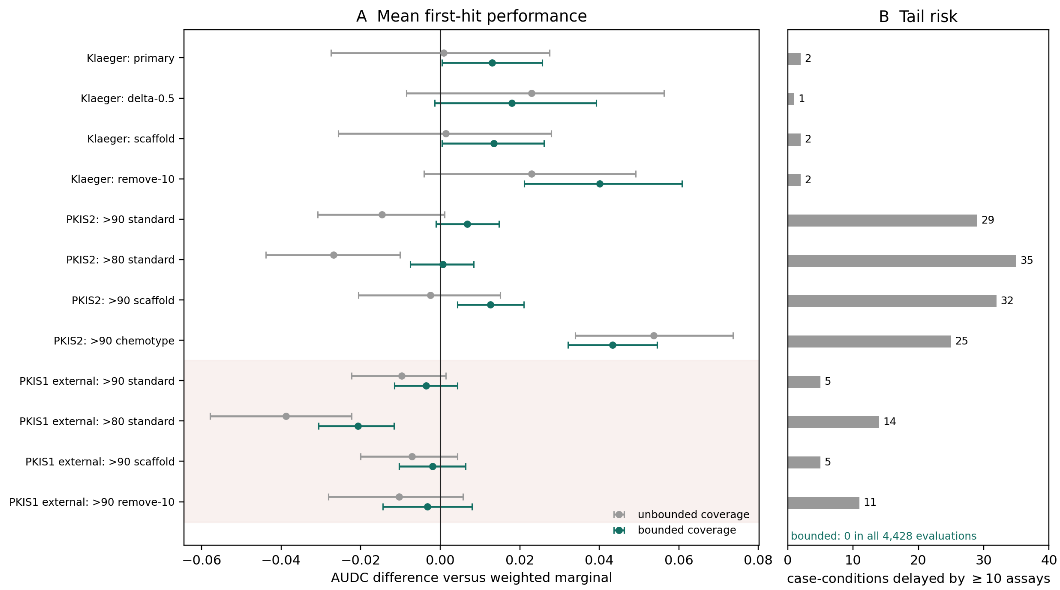

Figure 1: **A:** Paired mean AUDC effects with 95% bootstrap intervals. The shaded rows are the untouched PKIS1 external evaluation. **B:** unconstrained coverage creates large first-hit delays; the rank-deadline rule eliminates them by construction. Counts are compound–condition evaluations, not independent compounds.

Each compound is held out once. Conditions use all eligible references, remove exact Bemis–Murcko scaffold matches, remove ten nearest references, or—for PKIS2 only—remove source-chemotype matches. PKIS1 was selected and its method, four conditions, success criteria, and implementation hashes were frozen before any ranking or aggregate outcome. Two pre-outcome input stops documented raw missing cells and finite percent-inhibition values outside $[0, 100]$; missing cells remained missing and finite values were retained without clipping. A schema command had inadvertently printed two example processed rows before the freeze; they were not used to change any rule.

Metrics and inference. The primary outcome is area under the any-activity discovery curve (AUDC) over budgets 1–20. It is equivalent to a linear transform of censored cost to first hit. We also report hit rates at budgets 1, 3, 5, 10, and 20 and delays of at least ten assays. Intervals use 10,000 paired compound bootstraps; p-values use 100,000 paired sign flips, with Holm adjustment over the four external conditions. External success required nonnegative, noninferior (-0.015 margin) effects in all four conditions, a positive chemical-shift macro-effect with interval excluding zero, and zero large delays. The fixed 356-compound census gives an approximate 80%-power minimum detectable AUDC difference of 0.015–0.016 using development variances.

4 Results

Development evidence favors bounded diversification. The bounded component was prespecified before its run, although promoting it over a failed conformal wrapper occurred after results. Its effect versus marginal ranking was nonnegative in all eight development conditions (Fig. 1). Under PKIS2 chemotype exclusion, it gained 0.0433 AUDC (95% CI $[0.0321, 0.0545]$), retaining 80.6% of unbounded coverage's 0.0537 gain. PKIS2 scaffold exclusion gained 0.0127 ($[0.0043, 0.0210]$). Klaeger primary and remove-ten-nearest effects were 0.0131 ($[0.00045, 0.0257]$) and 0.0401 ($[0.0212, 0.0608]$), respectively.

The bound repaired the main downside of unconstrained coverage. Unbounded effects were negative for PKIS2 standard (-0.0146), > 80 (-0.0268), and scaffold (-0.0025), with 29, 35, and 32 large delays. Bounded effects were $+0.0068$, $+0.0006$, and $+0.0127$, with zero large delays. Nevertheless, bounded top-10 order stability under reference resampling was lower than unbounded coverage by 0.0251 ($[-0.0303, -0.0198]$); stability is not claimed.

Table 1: Untouched external PKIS1 effects. Δ is AUDC versus weighted marginal. Delay is bounded minus marginal censored assays (positive is worse). The frozen composite failed.

Condition	Δ bounded	95% CI	Δ unbounded	Mean delay
> 90 standard	$-.0035$	$[-.0115, .0044]$	$-.0097$	.070
> 80 standard	$-.0206$	$[-.0306, -.0117]$	$-.0388$	.413
> 90 scaffold	$-.0020$	$[-.0104, .0063]$	$-.0072$	.039
> 90 remove-10	$-.0032$	$[-.0145, .0080]$	$-.0104$	.065

The external mean benefit does not transfer. All four PKIS1 bounded estimates were negative (Table 1); the chemical-shift macro-effect was -0.0026 ($[-0.0113, 0.0061]$). At > 80 the negative effect was decisive (sign-flip $p = 10^{-5}$; four-condition Holm $p = 4 \times 10^{-5}$). Hence the external composite failed, and the data refute a universal improvement claim.

The risk result did transfer. Bounded coverage had no delay of ten or more in all 1,424 external evaluations; unbounded coverage had 35. Moreover, all four bounded-versus-unbounded point estimates were positive: $+0.0062$ AUDC at > 90 standard ($[0.0013, 0.0124]$) and $+0.0181$ at > 80 ($[0.0096, 0.0279]$). Reference-resampling stability was similar externally (-0.0023, $[-0.0097, 0.0052]$).

The failure is not a parent-collapse artifact. An explicitly post-result 18-cell audit retained the negative verdict. At the 224 assay-construct resolution, standard effects remained negative at > 80 (-0.0132, $[-0.0222, -0.0049]$) and > 90 (-0.0090, $[-0.0187, 0.0000]$). Parent-level standard effects were also negative at thresholds > 50, > 70, > 80, and > 90. Two exploratory remove-ten cells became positive at lower thresholds: > 50 gained 0.0281 ($[0.0111, 0.0449]$, Holm over 18 $p = 0.0141$) and > 70 gained 0.0205 ($[0.0059, 0.0354]$, Holm $p = 0.0968$). They define a possible regime hypothesis; they do not rescue the frozen external test.

5 Discussion

The clean result is a separation between a guarantee and a prediction. Under the weighted empirical-profile model, Eq. 1 values diversity because repeated assays on co-active profiles are redundant. That assumption was useful for PKIS2 chemotype shift and Klaeger, but chemical similarity did not make it transport to PKIS1. Rank deadlines cannot create transportability; they ensure that being wrong about it has a controlled first-hit consequence.

This distinction is useful for resource-constrained drug discovery. If a team has externally validated profile coverage for its assay platform and chemistry, bounded coverage preserves much of the diversity gain without the observed tail failures. Without such validation, marginal ranking remains the justified default. A pseudo-outcome gate and a label-free conformal-support wrapper were tested in the development sequence and did not solve this regime-selection problem; we keep them as negative ablations rather than present a menu of heuristics.

Important limitations remain. All evidence is retrospective. PKIS1 and PKIS2 are different compound panels but share the PKIS ecosystem; Klaeger adds a different binding modality but only 111 compounds. Percent-inhibition thresholds are operational labels, not potency estimates. Max-collapsing assay constructs changes the biological question, although the construct-level audit did not reverse the external result. Morgan similarity and reference profiles omit target structure, concentration response, cell context, pharmacokinetics, and assay cost. The method ranks kinase counter-screens; it neither predicts a therapeutic window nor recommends a compound or treatment.

A kinase-assay expert must review four points before submission or reuse: (i) parent-target collapse across mutants and complexes; (ii) comparability of Nanosyn percent inhibition, PKIS2 inhibition, and Kinobeads K_d while keeping their results separate; (iii) whether $> 80/ > 90$ and relative-K_d labels are appropriate for the stated counter-screen decision; and (iv) representative orders for known assay artifacts and promiscuous compounds. No such review is fabricated here.

6 Conclusion

Rank-deadline coverage gives an exact, inexpensive backstop for diversity-aware kinase assay ordering. It preserved meaningful development gains and removed all 163 large delays made by unconstrained coverage, but a frozen external panel rejected universal mean improvement. For AI-enabled drug discovery, that negative transfer is part of the contribution: a formal operational guarantee can survive where a retrospective performance story does not.

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A Reproducibility and data rights

The PKIS2 workbook is CC BY 4.0 and pinned by SHA-256 `48ead22a1...333e2a2c`. PKIS1 is a ChEMBL-derived raw export archived at repository commit `5fd3934f...1294122b`, pinned by SHA-256 `81d7f9f8...119c88a9`, and redistributed only under ChEMBL's CC BY-SA 3.0 terms. The Klaeger-derived ChEMBL 30 records are also CC BY-SA 3.0. The code records row-level provenance, versions, output hashes, and the chronology of protocol, two input amendments, implementation freeze, external result, and post-result robustness audit. Full runs are CPU-only, use no model training, and incurred USD 0 in paid API use.

B Development-model sequence

Unbounded profile coverage motivated the tail-risk criterion. A cross-fitted pseudo-outcome gate removed the PKIS2 > 80 reversal and reduced chemotype large delays from 25 to 10, but retained only 57.4% of the chemotype gain and reduced stability. A label-free conformal-support gate retained 78.3% and reduced the delay count to 21, missing the fixed limit by one and again reducing stability. The final conformal wrapper around bounded coverage retained 60.1% and failed its composite. The prespecified bounded component passed five of six development criteria; selecting it as the headline after observing that run is why PKIS1 was required.

C Biological-review handoff

The release includes a checklist with source-record examples for expert review. Any paper revision following expert feedback must distinguish corrections of biological interpretation from outcome-driven changes to the algorithm. A future decisive experiment is a preregistered prospective or time-split kinase panel in a discovery-relevant chemical series, with assay costs and confirmation measurements specified before ranking.